

TP n° 1

Préparation de l'environnement de travail
Base de données universitaire sous Oracle APEX
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1 Objectif

L'objectif de ce TP est de :

1. Se connecter à Oracle APEX (via le navigateur) ou à SQLPLUS (en ligne de commande).
2. Créer le schéma de la base de données universitaire.
3. Insérer les données de test.
4. Exécuter des requêtes de vérification.

Important Oracle : Utiliser VARCHAR2 au lieu de VARCHAR, et NUMBER au lieu de numeric. Oracle n'accepte pas le mot-clé numeric.

2 Connexion à Oracle APEX

Accéder à l'interface Oracle APEX via le navigateur :

```
https://<votre-instance>.oraclecloudapps.com/ords/sql-developer
```

Ou via SQLPLUS en ligne de commande :

```
sqlplus <username>/<password>@<host>:<port>/<service>
```

3 Suppression des tables existantes

```
1 BEGIN EXECUTE IMMEDIATE 'DROP TABLE prereq CASCADE CONSTRAINTS';
2   EXCEPTION WHEN OTHERS THEN NULL; END;
3 /
4 BEGIN EXECUTE IMMEDIATE 'DROP TABLE time_slot CASCADE CONSTRAINTS
5   ';
6   EXCEPTION WHEN OTHERS THEN NULL; END;
7 /
8 BEGIN EXECUTE IMMEDIATE 'DROP TABLE advisor CASCADE CONSTRAINTS';
9   EXCEPTION WHEN OTHERS THEN NULL; END;
10 /
11 BEGIN EXECUTE IMMEDIATE 'DROP TABLE takes CASCADE CONSTRAINTS';
12   EXCEPTION WHEN OTHERS THEN NULL; END;
13 /
14 BEGIN EXECUTE IMMEDIATE 'DROP TABLE teaches CASCADE CONSTRAINTS';
15   EXCEPTION WHEN OTHERS THEN NULL; END;
16 /
17 BEGIN EXECUTE IMMEDIATE 'DROP TABLE student CASCADE CONSTRAINTS';
18   EXCEPTION WHEN OTHERS THEN NULL; END;
19 /
20 BEGIN EXECUTE IMMEDIATE 'DROP TABLE section CASCADE CONSTRAINTS';
21   EXCEPTION WHEN OTHERS THEN NULL; END;
22 /
23 BEGIN EXECUTE IMMEDIATE 'DROP TABLE teacher CASCADE CONSTRAINTS';
24   EXCEPTION WHEN OTHERS THEN NULL; END;
```

```
24 /
25 BEGIN EXECUTE IMMEDIATE 'DROP TABLE course CASCADE CONSTRAINTS';
26     EXCEPTION WHEN OTHERS THEN NULL; END;
27 /
28 BEGIN EXECUTE IMMEDIATE 'DROP TABLE department CASCADE
29     CONSTRAINTS';
30     EXCEPTION WHEN OTHERS THEN NULL; END;
31 /
32 BEGIN EXECUTE IMMEDIATE 'DROP TABLE classroom CASCADE CONSTRAINTS
33     ';
34     EXCEPTION WHEN OTHERS THEN NULL; END;
35 /
```

4 Création des tables

```
1 CREATE TABLE classroom (
2     building    VARCHAR2(15),
3     room_number VARCHAR2(7),
4     capacity    NUMBER(4,0),
5     PRIMARY KEY (building, room_number)
6 );
7
8 CREATE TABLE department (
9     dept_name    VARCHAR2(20),
10    building     VARCHAR2(15),
11    budget       NUMBER(12,2),
12    PRIMARY KEY  (dept_name)
13 );
14
15 CREATE TABLE course (
16     course_id    VARCHAR2(8),
17     title        VARCHAR2(50),
18     dept_name    VARCHAR2(20),
19     credits      NUMBER(2,0),
20     PRIMARY KEY  (course_id),
21     FOREIGN KEY  (dept_name) REFERENCES department
22 );
23
24 CREATE TABLE teacher (
25     ID           VARCHAR2(5),
26     name         VARCHAR2(20),
27     dept_name    VARCHAR2(20),
28     salary       NUMBER(8,2),
29     PRIMARY KEY  (ID),
30     FOREIGN KEY  (dept_name) REFERENCES department
31 );
32
33 CREATE TABLE section (
34     course_id    VARCHAR2(8),
35     sec_id       VARCHAR2(8),
```

```
36     semester      VARCHAR2(6),
37     year           NUMBER(4,0),
38     building       VARCHAR2(15),
39     room_number    VARCHAR2(7),
40     time_slot_id   VARCHAR2(4),
41     PRIMARY KEY (course_id, sec_id, semester, year),
42     FOREIGN KEY (course_id) REFERENCES course,
43     FOREIGN KEY (building, room_number) REFERENCES classroom,
44     CONSTRAINT chk_semester CHECK (
45         semester IN ('Fall','Winter','Spring','Summer'))
46 );
47
48 CREATE TABLE teaches (
49     ID              VARCHAR2(5),
50     course_id       VARCHAR2(8),
51     sec_id          VARCHAR2(8),
52     semester        VARCHAR2(6),
53     year            NUMBER(4,0),
54     PRIMARY KEY (ID, course_id, sec_id, semester, year),
55     FOREIGN KEY (course_id, sec_id, semester, year) REFERENCES
56     section,
57     FOREIGN KEY (ID) REFERENCES teacher
58 );
59
60 CREATE TABLE student (
61     ID              VARCHAR2(5),
62     name            VARCHAR2(20),
63     dept_name       VARCHAR2(20),
64     tot_cred        NUMBER(3,0),
65     PRIMARY KEY (ID),
66     FOREIGN KEY (dept_name) REFERENCES department
67 );
68
69 CREATE TABLE takes (
70     ID              VARCHAR2(5),
71     course_id       VARCHAR2(8),
72     sec_id          VARCHAR2(8),
73     semester        VARCHAR2(6),
74     year            NUMBER(4,0),
75     grade           VARCHAR2(2),
76     PRIMARY KEY (ID, course_id, sec_id, semester, year),
77     FOREIGN KEY (course_id, sec_id, semester, year) REFERENCES
78     section,
79     FOREIGN KEY (ID) REFERENCES student
80 );
81
82 CREATE TABLE advisor (
83     s_ID            VARCHAR2(5),
84     i_ID            VARCHAR2(5),
85     PRIMARY KEY (s_ID, i_ID),
86     FOREIGN KEY (s_ID) REFERENCES teacher (ID),
87     FOREIGN KEY (i_ID) REFERENCES teacher (ID),
```

```

85     FOREIGN KEY (s_ID) REFERENCES student (ID)
86 );
87
88 CREATE TABLE time_slot (
89     time_slot_id VARCHAR2(4),
90     day          VARCHAR2(1),
91     start_hr     NUMBER(2),
92     start_min    NUMBER(2),
93     end_hr       NUMBER(2),
94     end_min      NUMBER(2),
95     PRIMARY KEY (time_slot_id, day, start_hr, start_min)
96 );
97
98 CREATE TABLE prereq (
99     course_id   VARCHAR2(8),
100    prereq_id    VARCHAR2(8),
101    PRIMARY KEY (course_id, prereq_id),
102    FOREIGN KEY (course_id) REFERENCES course,
103    FOREIGN KEY (prereq_id) REFERENCES course
104 );

```

5 Insertion des données

```

1  -- Classrooms
2  INSERT INTO classroom VALUES ('Packard','101',500);
3  INSERT INTO classroom VALUES ('Painter','514',10);
4  INSERT INTO classroom VALUES ('Taylor','3128',70);
5  INSERT INTO classroom VALUES ('Watson','100',30);
6  INSERT INTO classroom VALUES ('Watson','120',50);
7
8  -- Departments
9  INSERT INTO department VALUES ('Biology','Watson',90000);
10 INSERT INTO department VALUES ('Comp. Sci.','Taylor',100000);
11 INSERT INTO department VALUES ('Elec. Eng.','Taylor',85000);
12 INSERT INTO department VALUES ('Finance','Painter',120000);
13 INSERT INTO department VALUES ('History','Painter',50000);
14 INSERT INTO department VALUES ('Music','Packard',80000);
15 INSERT INTO department VALUES ('Physics','Watson',70000);
16
17 -- Courses
18 INSERT INTO course VALUES ('BI0-101','Intro. to Biology','Biology',4);
19 INSERT INTO course VALUES ('BI0-301','Genetics','Biology',4);
20 INSERT INTO course VALUES ('BI0-399','Computational Biology','Biology',3);
21 INSERT INTO course VALUES ('CS-101','Intro. to Computer Science','Comp. Sci.',4);
22 INSERT INTO course VALUES ('CS-190','Game Design','Comp. Sci.',4);
23 INSERT INTO course VALUES ('CS-315','Robotics','Comp. Sci.',3);

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24 INSERT INTO course VALUES ('CS-319','Image Processing','Comp. Sci
    .',3);
25 INSERT INTO course VALUES ('CS-347','Database System Concepts','
    Comp. Sci.',3);
26 INSERT INTO course VALUES ('EE-181','Intro. to Digital Systems','
    Elec. Eng.',3);
27 INSERT INTO course VALUES ('FIN-201','Investment Banking','
    Finance',3);
28 INSERT INTO course VALUES ('HIS-351','World History','History',3)
    ;
29 INSERT INTO course VALUES ('MU-199','Music Video Production','
    Music',3);
30 INSERT INTO course VALUES ('PHY-101','Physical Principles','
    Physics',4);
31
32 -- Teachers
33 INSERT INTO teacher VALUES ('10101','Srinivasan','Comp. Sci.'
    ,65000);
34 INSERT INTO teacher VALUES ('12121','Wu','Finance',90000);
35 INSERT INTO teacher VALUES ('15151','Mozart','Music',40000);
36 INSERT INTO teacher VALUES ('22222','Einstein','Physics',95000);
37 INSERT INTO teacher VALUES ('32343','El Said','History',60000);
38 INSERT INTO teacher VALUES ('33456','Gold','Physics',87000);
39 INSERT INTO teacher VALUES ('45565','Katz','Comp. Sci.',75000);
40 INSERT INTO teacher VALUES ('58583','Califieri','History',62000);
41 INSERT INTO teacher VALUES ('76543','Singh','Finance',80000);
42 INSERT INTO teacher VALUES ('76766','Crick','Biology',72000);
43 INSERT INTO teacher VALUES ('83821','Brandt','Comp. Sci.',92000);
44 INSERT INTO teacher VALUES ('98345','Kim','Elec. Eng.',80000);
45
46 -- Sections
47 INSERT INTO section VALUES ('BIO-101','1','Summer',2009,'Painter'
    , '514','B');
48 INSERT INTO section VALUES ('BIO-301','1','Summer',2010,'Painter'
    , '514','A');
49 INSERT INTO section VALUES ('CS-101','1','Fall',2009,'Packard','
    101','H');
50 INSERT INTO section VALUES ('CS-101','1','Spring',2010,'Packard',
    '101','F');
51 INSERT INTO section VALUES ('CS-190','1','Spring',2009,'Taylor','
    3128','E');
52 INSERT INTO section VALUES ('CS-190','2','Spring',2009,'Taylor','
    3128','A');
53 INSERT INTO section VALUES ('CS-315','1','Spring',2010,'Watson','
    120','D');
54 INSERT INTO section VALUES ('CS-319','1','Spring',2010,'Watson','
    100','B');
55 INSERT INTO section VALUES ('CS-319','2','Spring',2010,'Taylor','
    3128','C');
56 INSERT INTO section VALUES ('CS-347','1','Fall',2009,'Taylor','
    3128','A');

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57 INSERT INTO section VALUES ('EE-181', '1', 'Spring', 2009, 'Taylor', '
    3128', 'C');
58 INSERT INTO section VALUES ('FIN-201', '1', 'Spring', 2010, 'Packard'
    , '101', 'B');
59 INSERT INTO section VALUES ('HIS-351', '1', 'Spring', 2010, 'Painter'
    , '514', 'C');
60 INSERT INTO section VALUES ('MU-199', '1', 'Spring', 2010, 'Packard',
    '101', 'D');
61 INSERT INTO section VALUES ('PHY-101', '1', 'Fall', 2009, 'Watson', '
    100', 'A');
62
63 -- Teaches
64 INSERT INTO teaches VALUES ('10101', 'CS-101', '1', 'Fall', 2009);
65 INSERT INTO teaches VALUES ('10101', 'CS-315', '1', 'Spring', 2010);
66 INSERT INTO teaches VALUES ('10101', 'CS-347', '1', 'Fall', 2009);
67 INSERT INTO teaches VALUES ('12121', 'FIN-201', '1', 'Spring', 2010);
68 INSERT INTO teaches VALUES ('15151', 'MU-199', '1', 'Spring', 2010);
69 INSERT INTO teaches VALUES ('22222', 'PHY-101', '1', 'Fall', 2009);
70 INSERT INTO teaches VALUES ('32343', 'HIS-351', '1', 'Spring', 2010);
71 INSERT INTO teaches VALUES ('45565', 'CS-101', '1', 'Spring', 2010);
72 INSERT INTO teaches VALUES ('45565', 'CS-319', '1', 'Spring', 2010);
73 INSERT INTO teaches VALUES ('76766', 'BIO-101', '1', 'Summer', 2009);
74 INSERT INTO teaches VALUES ('76766', 'BIO-301', '1', 'Summer', 2010);
75 INSERT INTO teaches VALUES ('83821', 'CS-190', '1', 'Spring', 2009);
76 INSERT INTO teaches VALUES ('83821', 'CS-190', '2', 'Spring', 2009);
77 INSERT INTO teaches VALUES ('83821', 'CS-319', '2', 'Spring', 2010);
78 INSERT INTO teaches VALUES ('98345', 'EE-181', '1', 'Spring', 2009);
79
80 -- Students
81 INSERT INTO student VALUES ('00128', 'Zhang', 'Comp. Sci.', 102);
82 INSERT INTO student VALUES ('12345', 'Shankar', 'Comp. Sci.', 32);
83 INSERT INTO student VALUES ('19991', 'Brandt', 'History', 80);
84 INSERT INTO student VALUES ('23121', 'Chavez', 'Finance', 110);
85 INSERT INTO student VALUES ('44553', 'Peltier', 'Physics', 56);
86 INSERT INTO student VALUES ('45678', 'Levy', 'Physics', 46);
87 INSERT INTO student VALUES ('54321', 'Williams', 'Comp. Sci.', 54);
88 INSERT INTO student VALUES ('55739', 'Sanchez', 'Music', 38);
89 INSERT INTO student VALUES ('70557', 'Snow', 'Physics', 0);
90 INSERT INTO student VALUES ('76543', 'Brown', 'Comp. Sci.', 58);
91 INSERT INTO student VALUES ('76653', 'Aoi', 'Elec. Eng.', 60);
92 INSERT INTO student VALUES ('98765', 'Bourikas', 'Elec. Eng.', 98);
93 INSERT INTO student VALUES ('98988', 'Tanaka', 'Biology', 120);
94
95 -- Takes (selection)
96 INSERT INTO takes VALUES ('00128', 'CS-101', '1', 'Fall', 2009, 'A');
97 INSERT INTO takes VALUES ('12345', 'CS-101', '1', 'Fall', 2009, 'C');
98 INSERT INTO takes VALUES ('45678', 'CS-101', '1', 'Fall', 2009, 'F');
99 INSERT INTO takes VALUES ('98988', 'BIO-101', '1', 'Summer', 2009, 'A'
    );
100 INSERT INTO takes VALUES ('98988', 'BIO-301', '1', 'Summer', 2010,
    NULL);

```

```

101
102 -- Advisors
103 INSERT INTO advisor VALUES ('00128','45565');
104 INSERT INTO advisor VALUES ('12345','10101');
105 INSERT INTO advisor VALUES ('98988','76766');
106
107 -- Time Slots
108 INSERT INTO time_slot VALUES ('A','M',8,0,8,50);
109 INSERT INTO time_slot VALUES ('B','M',9,0,9,50);
110 INSERT INTO time_slot VALUES ('H','W',10,0,12,30);
111
112 -- Prerequisites
113 INSERT INTO prereq VALUES ('BIO-301','BIO-101');
114 INSERT INTO prereq VALUES ('CS-190','CS-101');
115 INSERT INTO prereq VALUES ('CS-347','CS-101');
116 INSERT INTO prereq VALUES ('EE-181','PHY-101');
117
118 COMMIT;

```

6 Vérification

```

1 SELECT 'classroom' AS table_name, COUNT(*) AS nb_rows FROM
   classroom
2 UNION ALL
3 SELECT 'department', COUNT(*) FROM department UNION ALL
4 SELECT 'course', COUNT(*) FROM course UNION ALL
5 SELECT 'teacher', COUNT(*) FROM teacher UNION ALL
6 SELECT 'section', COUNT(*) FROM section UNION ALL
7 SELECT 'student', COUNT(*) FROM student UNION ALL
8 SELECT 'takes', COUNT(*) FROM takes UNION ALL
9 SELECT 'prereq', COUNT(*) FROM prereq;

```

7 Schéma de la base de données

```

classroom(building*, room_number*, capacity)
department(dept_name*, building, budget)
course(course_id*, title, dept_name->, credits)
teacher(ID*, name, dept_name->, salary)
section(course_id*->, sec_id*, semester*, year*, building, room_number, time_slot_id)
teaches(ID*->, course_id*->, sec_id*->, semester*->, year*->)
student(ID*, name, dept_name->, tot_cred)
takes(ID*->, course_id*->, sec_id*->, semester*->, year*->, grade)
advisor(s_ID*->, i_ID->)
time_slot(time_slot_id*, day*, start_hr*, start_min*, end_hr, end_min)
prereq(course_id*->, prereq_id*->)

```

(*) = attribut de la clé primaire (→) = clé étrangère